

1. Playtest Index

One entry per run in docs/playtests/playtest_NNNN/ (see docs/skills/playtest.md §10 for the format and layout these follow). Newest first. Update this file whenever a new playtest is filed — that’s part of filing it, not a separate later chore.

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1.1. playtest_0011 — Vision Tooling, Client Configurations, Server BRP/MCP, and the Multi-Client Desync

Date	2026-09-24 17:20 – 20:30 UTC (one long session, several restarts)
Commit	f4180af “Fix no dev-tools feature build failure” at the end; the session’s work (server tools, ObserveRequest, no-render decorate fix, playbook updates) uncommitted as of this entry
Agent	opencode agent (GLM-5.3-Flash), with the project owner co-driving a windowed client
Report	docs/playtests/playtest_0011/report.typ

Three linked verifications on live sessions: the data-first vision tooling (game/ui, agent cursor, state-fused/cropped/unchanged-suppressed captures, 1280×800 viewport), the client-configuration matrix (rendered headless, --no-render, --headless-render observer, fleet ports) including a live two-client desync reproduction on full-render clients — second client spawns with its look pitch-pinned at the clamp, first client unaffected — and the new server-side BRP/MCP surface (server/state) whose very first use caught a client-vs-server divergence red-handed. Findings include the no-render second-player crash (found and fixed during the session), a measured zombie-player census, and the next-step instrumentation plan for the desync.

1.2. playtest_0010 — Optional CommonAssets Furniture / --no-common-assets

Date	2026-09-23 17:23 – 17:32 UTC
Commit	5e71215 “Implement Skein mesh primitives for agent testing”; the CommonAssets changes not yet committed as of this entry
Agent	opencode agent (GLM-5.3-Flash)
Report	docs/playtests/playtest_0010/report.typ

Implements playtest 0009’s closing proposal: the 9 engine-furniture fields of CommonAssets (fonts, WAVs, KTX2 skybox, atlas PNG pairs) became #[asset(key = "...", optional)] Option<Handle<T>> fields — a manifest that omits the keys resolves them to None and every consumer degrades gracefully (Bevy’s embedded default font, no skybox pass, no sample playback, an empty icon-atlas fallback) — plus a pre-sync --no-common-assets CLI flag for the even-bareer boot (no manifest read at all, a CommonAssets::placeholder() resource, and an immediate AssetLoading → MainMenu transition; the dev console’s hardcoded font path suppressed too). Verified in three modes: playtest 0009’s tree stripped **in place** to 100% plaintext (only text files remain, full loop works, clear color replaces the skybox), the flag mode (UI-only menu, in-game ClientWorldAsset visuals still render — that path loads by path, not manifest), and a production regression check (skybox renders, zero degradation warns). Also noted: the manifest’s lobby_background key is dead in every mode (the code aliases it to menu_background — pre-existing), and a pre-existing connect-time Disconnected re-entry into MainMenu observed in both modes.

1.3. playtest_0009 — Isolated, All-Plaintext Playtest Assets

Date	2026-09-23 16:00 – 16:40 UTC
Commit	5e71215 “Implement Skein mesh primitives for agent testing”
Agent	opencode agent (GLM-5.3-Flash)
Report	docs/playtests/playtest_0009/report.typ

First experiment with the owner’s isolated-asset idea: each playtest ships its own server/client assets under `playtest_assets/playtest_NNNN/` as **plaintext** — hand-written JSON .gltf scenes whose only content is Skein components (`ClientReplicate`, `ClientWorldAsset`, `ColliderConstructor`, and the new `MeshPrimitive` for zero-baked-data visuals), a per-playtest dynamic-asset manifest, config, and a single en-US locale. The whole get-in-game loop runs on the isolated assets: level list, colliders (the player grounds on the authored Skein collider), `MeshPrimitive` visuals rendering, starfield skybox, HUD — with the only non-text files being copied engine furniture (fonts, WAVs, KTX2, atlas PNGs). Two self-inflicted asset bugs found and documented en route (a glTF node-index typo; the pitch sign convention), plus a new playtest technique: injecting lights via `BRP.world.insert_resources`.

1.4. playtest_0008 — Full Keyboard+Mouse / Gamepad Reachability Sweep

Date	2026-09-23 13:35 – 14:09 UTC
Commit	Started at 736ec01; the <code>lobby.rs</code> fix not yet committed as of this entry
Agent	Claude (Sonnet 5)
Report	docs/playtests/playtest_0008/report.typ

Requested directly: verify every functional element (main menu, lobby, in-game) is reachable by both keyboard+mouse and gamepad, using only device-level input injection (no `game/trigger/` `game/select_level/` `game/` input shortcuts). Reproduced and root-caused a real, 100%-reproducible bug the project owner had independently noticed: a genuine mouse click on the lobby’s “Play” or “Main Menu” button never worked, at all — `client/src/ui/lobby.rs` imported the wrong of two identically-named `Activate` event types, so these `FeathersButton`-based handlers only ever received `Activate` via a gamepad/keyboard-Enter compatibility bridge meant for old-style widgets, never via a real click’s actual `bevy_ui_widgets::Activate`. Fixed with a one-line import change; verified post-fix with a fresh server+client pair that a mouse click on Play immediately after selecting a level (the exact reported scenario) now reaches `GameState::InGame` — and independently confirmed by the project owner on the real windowed client (Connect -> select level -> click Play, all with the mouse; literal Enter also works there). Also found and documented a separate, unfixable **harness** limitation: `bevy_input_focus::dispatch_focused_input` requires a `PrimaryWindow` entity that `-mcp` headless mode never creates, so literal keyboard Enter can never confirm a `FeathersButton` through this tool API (mouse click and gamepad South both work fine) — not believed to affect a real windowed client. Full reachability matrix in the report covers every menu/lobby/in-game control across all three input methods.

1.5. playtest_0007 — Add game/keyboard + game/mouse, Verify Real UI Clicks

Date	2026-09-23 08:33 – 08:41 UTC
Commit	Started at b340541; not yet committed as of this entry
Agent	Claude (Sonnet 5)
Report	docs/playtests/playtest_0007/report.typ

Extended `game/gamepad` (playtest_0006) to keyboard and mouse: `game/keyboard` mocks `ButtonInput<KeyCode>` directly (every one of Bevy’s 160+ variants, via `KeyCode`’s own `serde` impl); `game/mouse` mocks `ButtonInput<MouseButton>` plus drives `bevy_picking`’s real `PointerInput` pipeline for cursor motion/position/clicks — the first method in this API that reaches UI by screen position rather than by navigating focus and confirming. First implementation of mouse motion/wheel set `AccumulatedMouseMotion/AccumulatedMouseScroll` via a direct resource write — compiled and ran with no error, but silently did nothing (`look_yaw` stayed `0.0`), because Bevy’s own per-frame reset-from-events systems unconditionally overwrite those resources every frame. Fixed by injecting real `MouseMotion/MouseWheel` events instead. Verified live: a real mouse click on “Connect”/“Options” at their actual screenshot pixel coordinates drove the genuine `bevy_ui/bevy_picking` pipeline (state transitions and selector

popup open/close confirmed independently via game/state and log lines); KeyW moved the in-game player through the real replicated movement pipeline; mouse motion turned the camera only after the event-based fix.

1.6. playtest_0006 — Add game/gamepad, Verify the Literal Crash Path

Date	2026-09-23 01:37 – 01:42 UTC
Commit	Started at 05a416c; not yet committed as of this entry
Agent	Claude (Sonnet 5)
Report	docs/playtests/playtest_0006/report.typ

Added game/gamepad (mocks real bevy_input::gamepad::Gamepad button/axis state on a synthetic entity, flowing through bevy_enhanced_input's actual binding resolution) precisely to close playtest_0005's own stated gap: no way to reach UI navigation through the tool API, only the three ahoy gameplay actions via game/input's action-level mocking. First implementation used the wrong Gamepad field (digital, not analog – BEI's button reader calls Gamepad::get, which is the analog map; confirmed by testing, a silent no-crash failure, not an error) and was caught by actually running it, not by review. Fixed, then used immediately: drove the **literal** reported crash path for the first time – gamepad Start (open pause modal) -> DPadUp (navigate focus to “Main Menu”) -> South (activate) – confirming playtest_0005's fix against the real interaction, not an equivalent one. No crash, clean GameState::MainMenu transition.

1.7. playtest_0005 — Fix the return_to_main_menu Crash

Date	2026-09-23 01:22 – 01:24 UTC
Commit	Started at 80b6f0f; fix not yet committed as of this entry
Agent	Claude (Sonnet 5)
Report	docs/playtests/playtest_0005/report.typ

User-reported crash on the real windowed client: in-game -> pause modal -> “Main Menu” -> client panics. Matched a documented gap: return_to_main_menu was a live todo!(). Fixed to mirror lobby_main_menu_button's already-working pattern (commands.trigger(Disconnect); commands.set_state(GameState::MainMenu);), routing through the crate-local Disconnect event and its established two-part-disconnect handler rather than a fresh implementation. Verified via the equivalent code path (game/trigger disconnect while genuinely in-game) since the tool API has no way to simulate the actual keypress/button-click yet – no crash, clean return to MainMenu, UI correctly rendered. Also fixed a stale doc comment in modal_menu.rs caught along the way (no behavior change). **Re-verified against the literal crash path in playtest_0006**, once game/gamepad closed the tool-API gap this report's own verification had to work around.

1.8. playtest_0004 — Fix the Headless UI Render-Order Bug

Date	2026-09-23 00:53 UTC
Commit	Started at 41e29a9; fix not yet committed as of this entry
Agent	Claude (Sonnet 5)
Report	docs/playtests/playtest_0004/report.typ

Root-caused and fixed playtest_0003's open follow-on: the menu/lobby UI panel rendering **under** the background instead of on top of it. Real cause: bevy_ui's DefaultUiCamera::get() fallback only ever considers Window(Primary)-targeting cameras – structurally dead in --mcp mode, where everything targets Image – so headless mode's permanent bootstrap UI camera and player_camera()'s own self-tagged IsDefaultUiCamera collided the moment a player existed, breaking bevy_ui's camera selection entirely (not an ordering problem, despite looking like one). Very likely explains why all three of playtest_0003's ordering-only fix attempts regressed in-game rendering in ways that resisted explanation at the time. Fixed by actively maintaining “exactly one live IsDefaultUiCamera holder” as an invariant, handed off between the bootstrap camera and player_camera() as they come and go. Verified end-to-end: main menu, lobby, in-game (HUD now renders too, not just the world), and – newly tested, not covered by any earlier playtest – the full disconnect-back-to-main-menu round-trip all composite correctly.

1.9. playtest_0003 — Diagnose and Fix the Headless Camera Rendering Bug

Date	2026-09-22 23:00 – 2026-09-23 00:15 UTC
Commit	Started at 329c9bc; fix landed as e5fe6c6 (committed after this run’s verification) – spans both sides of a code change, not a static state
Agent	Claude (Sonnet 5)
Report	docs/playtests/playtest_0003/report.typ

Root-caused and fixed the bug playtest_0001/playtest_0002 both hit: -mcp mode’s cameras never rendered anything, because `bevy_render::camera::camera_system`’s `target_info` recompute silently never fires for a camera retargeted after Startup (a one-line fix, `projection.set_changed()`, in `retarget_cameras_to_offscreen`). This also explained the previously-separate “KTX2 skybox kills offscreen rendering” finding – same root cause, not a real skybox bug; the skybox/TAA/SSAO strip workaround is removed. Surfaced a follow-on UI-panel-renders-under-the-background ordering bug in menu/lobby specifically; three fix attempts were tried and reverted here (each one regressed in-game rendering worse than the ordering bug itself) – see the report’s “A follow-on issue found, attempted, and reverted” section for what was tried and why. **Fixed in playtest_0004**, once the real (non-ordering) cause was found. Also found, unrelated: the “minimal” level has no light source anywhere in its content (renders black on any client, not a -mcp-specific issue).

1.10. playtest_0002 — Bug Reproduction Pass (Caster-Resolution + Headless Camera)

Date	2026-09-22 23:25 UTC
Commit	329c9bc “Add playtest.md skill”
Agent	Claude (Sonnet 5)
Report	docs/playtests/playtest_0002/report.typ

A targeted re-verification pass, not a full state tour: checked two things `AGENTS.md`’s gap list already claimed, with harder evidence than playtest_0001 had supplied for the same claims. Confirmed live: `spawn_cube` is silently dropped server-side (zero Cube entities before/after the trigger, total server-log silence) – **still open, not fixed by any playtest since**, see `AGENTS.md`’s caster-resolution gap entry for the fix shape. Also found a sharper diagnostic lead for the headless camera bug (`Camera.computed.target_info` staying `null` specifically for cameras claimed after Startup) that playtest_0003 went on to root-cause and fix.

1.11. playtest_0001 — Headless (–mcp) Client State Tour & Input Drive

Date	2026-09-23 02:38 local (2026-09-22 22:38 UTC)
Commit	89f5eca “Implement headless –mcp mode for client”
Agent	opencode session, GLM-5.3-Flash
Report	docs/playtests/playtest_0001/report.typ

The first formal playtest: verified the agent/QA tool API (ADR 0009) end-to-end against a fresh server and a fresh headless client – drove every reachable client state, injected input through the real replicated pipeline, and recorded what the agent sees vs. what a human on a windowed client sees. Confirmed the movement/input/replication data path is fully solid headless (server- authoritative sim, client prediction, all verified through state reads across a full menu→game flow). First recorded the in-game blank-rendering bug and the KTX2-skybox-kill finding – both later found to be the same root cause, fixed in playtest_0003.